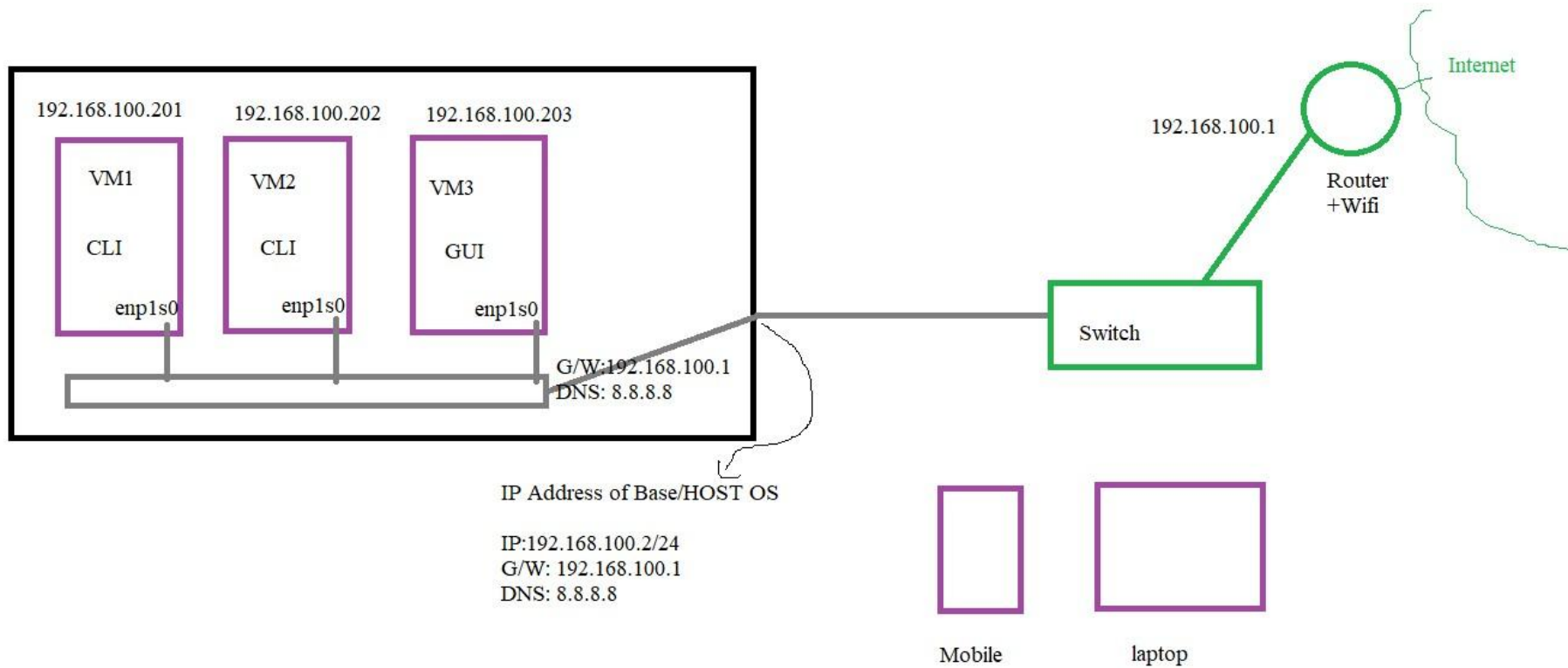


Hi Please Find the answers of the below question and submit the answers asap.

Topology



Basic – File Management:

Q1. Create a user named student and set password as student;

Q2. Create Directory and file as student user

Create a directory name "backup" on students home directory

In the above directory

Create 2 files named filea & fileb with content "hello happy learning"

Create some files with the name “songs1.mp3, songs2.mp3, songs3.mp3 to songs99.mp3”

Create some files with the name “film1.avi, film2.avi to film10.avi”

Create a some directory under /home/student/backup/ with the name songs, movies, picture, data

Create a some directory under /home/student/backup/ with the name dir1, dir2 to dir20

Q3: As student user

Move the files “songs1.mp3, songs2.mp3, songs3.mp3 ... songs99.mp3” to /home/student/backup/songs/

Move the files “film1.avi, film2.avi to film10.avi” to /home/student/backup/movies/

copy /etc/ directory to /home/student/backup/data/

Q4. List the students users home directory in tree format

Q5. Check the file size and inod number of file

/home/student/backup/filea &

/home/student/backup/fileb

Q6 Create a directory & file as root user

Create a directory under /tmp/ with the name r-dir1

Create a file under /tmp/r-dir1/root-file1 with content "hello its root users file"

Q7: Check the directory size of

`/home/student/backup/`

Q8 As student user

Copy the directory `"/home/student/backup/"` to

student's home directory with the name `"stud-backup1-dir1"` &

copy to `/tmp/stud-backup2-dir2`

Note: make sure permission, owner and group are preserved

Q9. Move the directory `/home/student/stud-backup1-dir1` to `/tmp`

List the Directory in `/tmp` and `/home/student` in tree format

Q10. Create the hard link for the file & directory

`/tmp/stud-backup1-dir1/filea` to students home directory

`/tmp/stud-backup1-dir1/` to students home directory

Q11. Delete the directory /tmp/stud-backup1-dir1/ and check whether you are able to access the content

/home/student/filea

/home/student/stud-backup1-dir1

Q12: As student user

copy last 5 lines from /etc/passwd to /tmp/my-passwd

Q13. As student user

List the /etc/ directory and copy the output to the file /tmp/etc-out.txt

Q14. As student user

Find all files named passwd by user student and direct the output to file /tmp/out2.txt and error on /tmp/err.txt

Q15 As root user

Find the keyword named err in /var/log/message and direct the output to /tmp/out3.txt

As root user

Find all the files in /etc/ directory which has the content "student" -----

Q16 find the no of lines with the keyword err in /var/log/message

Q17 find the no of files in /etc/ directory with the file name yum

Q18. Add a variable for the user student

variable named tmp_{list}=/tmp/*

variable named FN=James

variable named LN=Bond

Set Variable FN as environment variable

Q19. Check all the variables are listed

List file using the variable using “tmp_{list}”

Q20. Create a directory /mypgm1 & a create a script file under /mypgm1/myscript

File should contain the following content

```
echo "welcome to the pgm"
echo
echo todays date is $(date +%x)
echo
echo Partion info is
echo
lsblk
echo
echo Memory details are
echo
free -h
echo
echo processor details are
echo
cat /proc/cpuinfo
```

when u run the above file, it should display the output of the command mentioned in the file

Q21. File/script under /mypgm1/myscript should be executed as a command “myscript” by all the user

In future any script file or pgm which goes under /mypgm1/ should be taken up as command for all the users Q22: Find the location of the command “date”

Q23: copy the file /usr/bin/date to /mypgm1/samay

Q24: check if you are able to run the command samay as student user and root user

Managing Local Users and Groups

Q25. Create a user name & Group

mohanlal, mamootty belongs to group actor and supplementary group mollywood, bollywood nivin belongs to group actor and supplementary group mollywood sharuk, salman belongs to group actor and supplementary group Bollywood johny belongs to group actor and supplementary group Hollywood, comment filed should be “Johny Depp”

Password for the above user should be redhat

All the users password should be expired every 30 days

All the users password should be forcefully reset on next login

Verify which all group belong to the user mohanlal

Verify which all users belong to group “actor”

Q26. user mohanlal, mamootty should have admin rights.

Do some admin task, like: (Verify if you can view the content /etc/shadow and create a user name “bob” using any of the above user)

Q27. Create a user eric with used id 3345.

Q28: Create a group named tollywood with the id 5555

Q29 user eric’s account should be expired after 60days and

user mohanlal account should expire on 2022 dec 31

Q30. user sharuk should not have login shell

Q31. Lock user salman

Q32. Create a system user named app1 and the user should not have login shell

Q33. ADD USERS.

Create the following users:

Create a group "adminuser"

Create a user natasha who has "adminuser" as a supplementary group.

Create a user harry who also has "adminuser" as his supplementary group.

Create a user sara who does not have an interactive shell nor does belong to the group "adminuser"

All users should be set with a password flectrag

Controlling Access to Files

Q34. change the permission of the /tmp/stud-backup2-dir2

owner should be eric and group should be manager

owner should read and write group should read

and other should not have any rights

Q35. CREATE DIRECTORY.

Create a directory /data/dir1/

The user natasha and harry should be able to collaboratively work on this directory.

The files and directories created within this directory should automatically belong to the group adminuser.

All members of the group should have read and write access.

All other users should not have any permission.

Note: By default, root user will have read and write access on all files and directories.

Q36. Only owners & root should be able to delete files under the directory /data/dir1/

Q37. The default permission of the user mohanlal who creates file should be 660

Q38. Create the /etc/profile.d/my-umask.sh file with the following content

to set the umask to 007

for users with a UID greater than 199 and with a username and primary group name that match,

and to 022 for everyone else:

Q39: File /tmp/r-dir1/root-file1 shouldn't be deleted by any user including root (Addon)

Q40. create a file as root user under /tmp with a name root-filea (Addon)

File should be edited by user natasha and harry

All other users should not have any permission on this file

Q41. As student user (Addon)

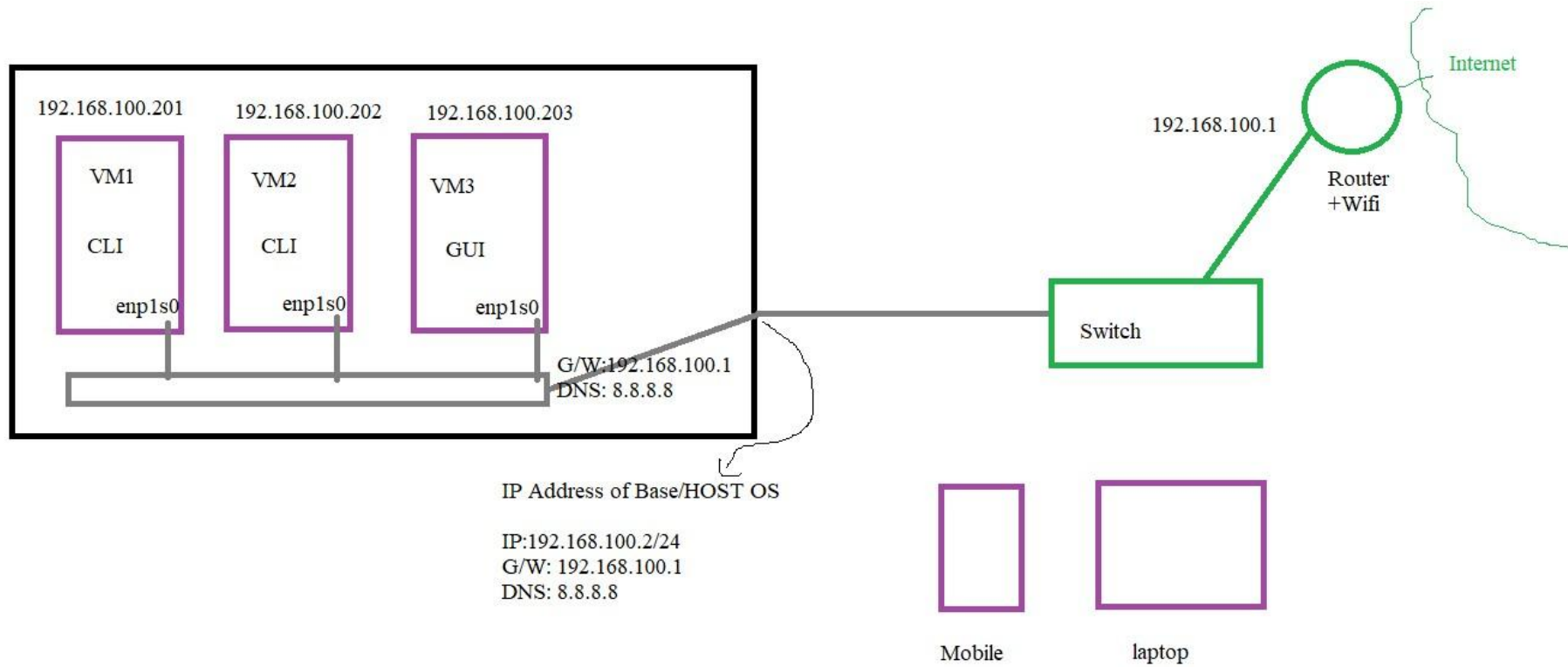
Create a file test.txt, and then delete the file test.txt

You might have noted that, it's not prompting to delete the file

Now configure things such a way that when user student deleted files it should prompt for confirmation

Managing Networking

Q42. Configure Network



Configure IP address on your VM's (VM1, VM2, VM3)

Requirement:

- VM's within the Host PC should be able to communicate with the other VM's
- VM's within the Host PC should be able to communicate with Host PC
- VM's within the Host PC should be able to communicate with the laptop and Mobile
- VM's within the Host PC should be able to communicate with Internet.
- Host PC should communicate with the VM's
- PC's & Mobile outside the Host PC's should communicate with the VMS

On VM1: Set Static Address Set IP: 192.168.100.201 SNM: 255.255.255.0 Gateway: 192.168.100.1 DNS: 8.8.8.8 Connection name: home1	On VM2: Set Static Address Set IP: 192.168.100.202 SNM: 255.255.255.0 Gateway: 192.168.100.1 DNS: 8.8.8.8 Connection name: home1	On VM3: Address Should be received from Router (DHCP) Set DNS as DNS: 8.8.8.8 Connection name: home1
--	--	---

Note: You can replace 192.168.100.X with the IP range of your local network

From VM's:

- List Connection name: home1
- Check the IP address, SNM, Gateway, DNS
- Check the active or listening N/W port
- Check the Support Network speed, current speed and Duplex

Q43: Configure hostname

VM1: servera.lab.example.com

VM2: serverb.lab.example.com

VM3: workstation.lab.example.com

Q44.

VM1's should communicate with each other using domain name and ip address.

For example, from VM1, VM should communicate with VM2 with serverb.lab.example.com & serverb Optional:

Q45 From servera, Show all the hops between the servera and google.com.

Q46. Check current established connections

Q47. check, which all networking service's/ports are opened(listening to)

Q48. On servera add secondary address in the connection "home1" and set the ip address to 192.168.200.101,subnet mask 255.255.255.0

Q49. On serverb add secondary address in the connection "home1" and set the ip address to 192.168.200.102,subnet mask 255.255.255.0

Q50. Verify the secondary address is added

Q51. Check the connectivity of servera from serverb with the new secondary address and vice versa.

Remote Access

Q52. Configure cockpit on VM1 , VM2 (If RHEL)

Q53. Make sure that servera, serverb access be accessed using ssh from workstation.

Q54. Create a user named “devops” in servera and serverb and the user should be accessed from workstation as student user without prompting the password.

Q55. Prohibit the root user form logging using ssh on servera & serverb.

Q56. Check who all have accessed ssh on servera and serverb for the past few days.

Q57. Create a user named user1 using cockpit service

Q58. Create a user name tom on servera and user name jerry on serverb. Password for the 2 users should be redhat

Q59. tom user from servera should be able to access user jerry on serverb without prompting for password

Service:

Q.60 On workstation stop ssh service and ensure ssh doesnt start automatically after reboot and also ensure that it cannot be started using "systemctl start sshd"

Q61. On workstation disable gui and system should be booted with CLI

Process management:

Q62. Create a script called mypgm in the /home/student/bin/ directory with following content

```
#!/bin/bash  
  
while true; do  
    echo -n "$@" >> /tmp/output_file  
    sleep 2  
done
```

Run the script as background process by passing the argument test1 for 60 seconds

Run the script again as background process by passing the argument test2 for 60 seconds

After running check the file /tmp/output_file, should have the content test1 and test2

Stop the above process after 60 sec and then continue it after 2 min for another 60 sec

Then kill both the process

Q63.

Create and run the below script in the background with the name monitor under the /home/student/bin directory with the following content:

```
#!/bin/bash
```

```
LOGFILE=/var/tmp/system-monitor.log
```

```
while true
do
{
echo "===== $(date) ====="
uptime
free -h
df -h
ps -eo pid,user,%cpu,%mem,cmd --sort=-%cpu | head -10
echo
} >> "$LOGFILE"

sleep 10
done
```

Identify the process using top and kill

Q64. check the architecture of cpu

check the operating systems version

check the ram size

Q65.

system should have to recommend profile

set it as the default profile

Q66.

Use a looping command to start multiple instances of the "sha1sum /dev/zero &" command.

Start two per virtual processor. For example if your system has 2 processors then 4 instance of the above process should be executed.

Identify the process and kill

Q67.

Run firefox in the backgroud on workstation with high priority

Partition:

Q68. Add 3 storage in servera

HDD1 5GB

HDD2 7GB

HDD3 8GB

Q69.

On servera create an msdos/mbr partion with 1G on HDD1

File system should be vfat and mounted on /mnt/songs

Mount should be persistent

Q70.

On servera create an GPT partion with 1.5G on HDD2

File sytem should be ext4 and mounted on /mnt/movies

Mount should be persistant

Q71. Create a swap partition on servera on HDD2

Size should be twice the size of RAM

swap should be persitant

Analyse Log:

Q72. Configure log server in such a way that all the info log is stored in /var/log/ccoeinfo

Q73. Test your log server, by directing a message called testing to the file /var/log/ccoeinfo

Q74. Find all the logs generated by the service sshd.service

Q75. Find all the error logs

Q76. Check the last 50 generated logs

Q77. Find all the logs generated by user root

Q78. Preserve the system journal logs

Q79. Configure servera as NTP server

Q80. Make sure the timezone is set to Asia/Kolkata

Q81. ON workstation, time should be synchronised from servera.lab.example.com and time.google.com

Package Installation

Q82. On servera/serverb/workstaion make sure packages can be installed

Configure repository from DVD/ISO or from internet.

Q83 On servera - Install vsftpd pakage

Q84 On servera - Install httpd package

Q85 On servera - Install the group package "Development Tools"

Q86 On servera - Install the **python36** module from the 3.6 stream and the common profile. Verify the current status of the module.

Q87 On servera - Install the **postgresql** module from the 9.6 stream and the server profile. Verify the current status of the module.

Q88. On servera - Add EPEL repository for additional packages

Q89. Install Ansible package

Q90. Install Libre office

Q91. Install chrome

Future Job:

Q92.1 CRON JOB.

natasha must run a job "logger testing" every 2 minutes

Q92.2 CRON JOB.

Every Tuesday at **1 PM**

Remove:

- unused images
- unused build cache

Archiving and Copying Files Between Systems

Q93. BACKUP FILES.

Create an archive /root/new.tar.gz which stores the backup of /usr/local.

Q94. On workstation create a directory under /backup

Copy /var/log directory from servera and serverb to workstation under the directory /backup.

This should happen on every weekend at 9:30 pm.

ACL

Q96. COPY FILE.

Copy the file /etc/fstab to /var/tmp/fstab .

The file should belong to the user root and group root.

The user natasha should be able to read and write on the file.

The user harry should neither read nor write on the file.

All other users should have read permission on the file.

Search Files

Q97 FIND FILES.

Find all files owned by the user era and copy it to /root/findfiles.

Q98 SEARCH WORDS.

Display the matches for the words "seismic" in the /usr/share/dict/words and save the output to a file /root/wordlist.

Script

Q99 Create a Script.

Create a script under /root/myprogram2

Create a script so that when you pass the argument redhat, the output should be displayed as fedora

when you pass the argument rhcsa, the output should be displayed as rhce

when argument no argument is passed , it should display "enter redhat or rhcsa"

Q100 Script to FIND FILES-2.

Create a script under /root/myprogram1

Find all files from /usr/ who's size is greater than 30K and less than 50K and copy it to /root/findfiles-2.

Also the files should have SUID as the permission

Q101 Write a script in serverb

Create a user called ashiq

Set password

Set account expiry after 60days

Install webserver

Start the service

Add firewall rules

Create index.html file with content testing

Ashiq user should be able to edit index.html files or can be able to create files under the doc root of web server

SELINUX

Q102 Configure servera with enforcing mode

Q103 Install web server on serverb

- Configure Apache
- Create the new document root to , **/custom**
- Add html pages
- Web server should be accessible from outside

Q104 user home directory should be accessible by web server

Q105 Web server should run with 82 port number and it should be accessible from outside

Managing Network Security

Q106 serverb should only be access from servera and workstation

Implementing Advanced Storage Features

Q107 LVM CREATION on servera.

Create a logical volume 50 extents where one extend having the size of 16MiB.

The logical volume has the name of database and volume group have the name of datastore.

The logical volume should be mounted under the directory /mnt/database with a file system of ext4 and should be automatically available on reboot.

Create a file name file1 on /mnt/database/

Q108 LVRESIZE.

Resize the logical volume database to 1500M which belongs to the volume group 'datastore'. Any size between 1400M to 1600M is permissible.

Create a file name file1 on /mnt/database/

NFS

Q111 Storage.

servera.lab.example.com

On servera.lab.example.com share a directory /data/rhome/remotouser1 using NFS. Share directory should be accessible from everyone

On workstation:

Create a user name remotouser1 and home directory should be /rhome/remotouser1

Password of the user 'remotouser1' is flectrag.

The 'remotouser1' home directory should be automounted locally beneath /rhome/remotouserx from servera:
/data/rhome/user1

The home directories must be writable by their users.

Running Containers

Q112 CONFIGURE CONTAINERS.

Create a container logserver from an image rsyslog in servera, from registry.redhat.io
login with registry.redhat.io with the redhat.com account
Configure the container with systemd services by an existing user "tommy"

Service name should be container-logserver and configure it to start automatically across reboot

Configure you host journal to store all journal accrss reboot

Copy al *.journal from /var/log/journal and all subdirecotries to /home/ tommy/container_logserver configure to automount
/var/log/journal from logserver (container) to /home/ tommy/container_logserver when container start.

LEVEL 2

In Ubuntu

- Update and upgrade the system. Remove unused packages.
- Configure UFW for SSH, HTTP and HTTPS. Allow SSH only from a specific subnet.
- Create user:
 - Username: appuser
 - Password: appuser
 - Home directory: /app/appuser

- Using user appuser:
 - Run nginx container named my-app1 on port 8082.
 - Content:
 - For nginx - Host a index.html page with content "Welcome to PDD"
 - Store content in /app/appuser/data
 - For mentioned container, data should be persistent. You can use the local storage or NFS configured earlier in one of the previous task.
 - Deploy PostgreSQL using Docker Compose.
 - Configure Apache or Nginx as a reverse proxy to access the container “my-app1”.
 - Configure HTTPS. with domain eg: pddtestapp.luluone.com - Create a self-signed Certificate or do inform us and we shall share the Certificate.

- Disk maintenance:
 - Automatically delete files larger than 500MB daily under /app/appuser/data
- Create cron job (2 AM daily):
 - Backup /var/log to /backup
 - Create tar archive with date